

Editorial

Large language models in biomedicine and health: current research landscape and future directions

Large language models (LLMs) are a specialized type of generative artificial intelligence (AI) focused on generating natural language text. These models are developed through extensive training on massive amounts of text data and use deep learning algorithms to generate new text that closely resembles human-generated text. Generative AI methods, including LLMs, are rapidly transforming various domains, including biomedicine and healthcare.^{1–6} They have already demonstrated remarkable potential as a means to process and analyze large amounts of text, interpret natural language, and generate new content in these domains. For example, Nori et al reported that GPT-4 is able to correctly answer the majority of questions from medical practice licensing exams, comfortably obtaining a passing grade.⁷ Similarly, Stribling et al found that this model exceeded the average performance of students in the graduate medical sciences on the majority of examinations, including strong performance on short answer and essay questions.⁸ Even though passing the exam is not the same as applying the knowledge in a real-world setting, these results demonstrate that LLMs can generate appropriate multiple-choice and narrative responses to questions framed in natural language.

ChatGPT, first released in November 2022, has garnered phenomenal attention from both the scientific community and a broader society. A keyword search of “large language models” OR “ChatGPT” in PubMed returned over 4500 articles that discuss the technology and its implications for various topics, including medical informatics, by the end of June 2024. In addition, LLM-based technologies have already been deployed in several healthcare systems and are offered as integrated products for use in the clinic within vendor electronic health record systems (for thoughts on initial evaluations of an early product, see Garcia et al⁹ and Tai-Seale et al¹⁰). This rapid adoption of LLMs like ChatGPT brings an unprecedented opportunity to use this novel AI technology to transform healthcare and medicine.

Despite their potential benefits, LLMs can sometimes produce invalid and unsubstantiated responses, a phenomenon known as the “hallucination and confabulation issue” in the literature, or biased responses, due to the biases inherent in their training data.^{11–17} With this great potential also comes the need for trustworthy and responsible development and use of technology. As we continue to explore the capabilities of ChatGPT and other LLMs, it is critical to address related ethical, legal, and social issues to ensure that the technology is used in ways that are safe, fair, trustworthy, and beneficial for all. In the

context of biomedicine and healthcare, it is particularly important to engage stakeholders, such as AI researchers, developers of data-driven clinical decision support, care providers, and system implementers from both academic medical centers and industry, to ensure responsible use of LLMs for good.

To accelerate research and development in this area, we issued a call for submissions in Summer 2023, specifically focusing on the intersection of biomedicine/health and LLMs, and invited contributions on all related aspects. We invited submissions that report on innovative informatics methods development and evaluation, as well as studies that demonstrate the effectiveness/limitations of LLMs methodologies in healthcare. We particularly encouraged submissions that address the challenges and opportunities of this intersection and offer new insights into how these fields can work together to advance healthcare.

This editorial provides an overview of the papers accepted in this Focus Issue. We highlight major themes and unique aspects of the research papers in medical LLMs, discuss ongoing challenges, and recommend future research directions. [Box 1](#) lists the relevant large language model terms and abbreviations used in this editorial.

Overall statistics of the Focus Issue

This JAMIA Focus Issue on LLMs in biomedicine and health has drawn enthusiasm from many researchers across different research disciplines. In total, we received over 150 submissions from authors in 25 countries and regions across 6 continents worldwide. The rigorous JAMIA peer review process was applied to all submissions, 41 of which were ultimately accepted for publication in the Focus Issue ([Table 1](#)). The majority of the accepted papers were authored by authors in North America, followed by those in Asia and Europe ([Figure 1A](#)).

The Focus Issue highlights the nature of multi-disciplinary collaboration in medical informatics research across the broad JAMIA community. The number of authors per paper varies from 1 to 23, with an average of 7.3. Many papers feature authors with diverse expertise from different departments and organizations. The authors’ expertise spans a wide range of fields, including computer science, data science, informatics, statistics, medicine, nursing, clinical services, public health policies, and more. Several papers also demonstrate scientific collaborations across different sectors, including academia, government labs, research institutes, hospitals, and industry. Additionally, a few papers showcase international collaborations among authors.

Box 1. Large language model terms and abbreviations.**Large language model terms**

Large language models: A large language model is a type of artificial intelligence designed to process and generate human-like text. A large language model is built using deep learning techniques, particularly neural networks with many layers, and is trained on text corpora to perform tasks such as language translation, summarization, and question answering.

Prompt engineering: Prompt engineering involves crafting inputs or “prompts” that guide large language models to generate desired outputs without changing their parameters. Effective prompt design can significantly enhance the performance of LLMs.

In-context learning: In-context learning refers to model learning to do a downstream task simply by conditioning on a prompt consisting of input-output examples.

Zero-shot learning: Zero-shot learning is a problem setup in deep learning where, at test time, a learner observes samples from classes which were not observed during training and needs to predict the class that they belong to. In the context of LLMs, this involves providing no specific examples in the prompt to guide the model for generating output.

One-shot learning: One-shot learning aims to classify objects from one example. In the context of LLMs, this involves providing only one specific example in the prompt to guide the model for generating output.

Few-shot learning: Few-shot learning refers to the ability of a model to learn a new task from a very limited number of examples (shots). In the context of LLMs, this involves providing a few specific examples during the prompt to guide the model on how to handle similar tasks, improving its accuracy and adaptability.

Chain-of-thought prompting: Chain-of-thought prompting is a technique used with large language models to encourage step-by-step reasoning in their responses. By explicitly asking the model to think through the steps of a problem, it can provide more transparent and logically structured solutions.

Retrieval-augmented generation: Retrieval-augmented generation combines traditional language modeling with a retrieval component that searches for relevant information. This enhances the model’s responses by grounding them in factual data.

Fine-tuning: Fine-tuning is an approach to transfer learning in which pre-trained model parameters are trained on a new dataset.

Instruction tuning: Instruction tuning is a technique for fine-tuning large language models on a labeled dataset of instructional prompts and corresponding outputs. It improves model performance not only on specific tasks but also on following instructions in general, thus helping adapt pre-trained models for practical use.

LoRA fine-tuning: LoRA fine-tuning is a technique that accelerates the fine-tuning of large models while consuming less memory. It uses 2 smaller matrices (the LoRA adapters) through low-rank decomposition for weight updates while keeping the LLMs’ original weight matrix frozen at fine-tuning, then combine the updated adapter weights with the LLMs’ original weights for inference.

QLoRA fine-tuning: QLoRA fine-tuning represents a more memory-efficient iteration of LoRA fine-tuning. It takes a step further by quantizing the weights of the LoRA adapters (the smaller matrices) to lower precision (eg, 4-bit instead of 8-bit).

HIPAA-compliance: HIPAA-compliant systems adhere to HIPAA (Health Insurance Portability and Accountability Act) regulations, ensuring they meet the required standards to protect patient health information. HIPAA sets national standards for the protection of health information in the United States. It ensures the privacy and security of individually identifiable health information.

Abbreviations

AI: Artificial intelligence

AUROC: Area under the receiver operating characteristic

AUPRC: Area under the precision-recall curve

BioNLP: Biomedical natural language processing

ED: Emergency department

Exp. CLA: Explicit criterion-level accuracy

FP: False positive

FPR: False-positive rate

IE: Information extraction

Imp. CLA: Implicit criterion-level accuracy

IR: Information retrieval

L_Rel: Last relevant document

LLMs: Large language models

LoRA: Low-rank adaptation

MAP: Mean average precision

MCQ: Multiple-choice question

ML: Machine learning

NDCG@10: Normalized discounted cumulative gain at the 10th item

NER: Named entity recognition

P@10: Precision at the 10th item

PPV: Positive predictive values

QA: Question answering

QLoRA: Quantized low-rank adaptation

R@k%: Recall at k %

RAG: Retrieval-augmented generation

RE: Relation extraction

RWD: Real-world data

SLMs: Small language models

TCM: Traditional Chinese medicine

TP: True positive

WSS@r%: Work saved over sampling at an r % recall level

Table 1. List of accepted papers and their tasks, large language models, datasets, and evaluation methods.

Paper Ref.	Task	Large language models	Dataset	Evaluation (automatic vs human)
Research and application articles				
Acharya et al ¹⁸	Clinical risk prediction	Med-Bert, Llama 2	SIPPS, MIMIC-IV, Synthea	Auto (AUROC, AUPRC)
Akinseloyin et al ¹⁹	Prioritization of candidate studies for systematic review	GPT-3.5	CLEF eHealth 2019 Task 2	Auto (L_Rel, MAP, R@k%, WSS@r%)
Andreadis et al ²⁰	Assessment of demographic bias in clinical recommendations	GPT-4	Private	Auto (accuracy) + Human
Bak and Chin ²¹	Identification of motivation states, predict behavior change	ChatGPT, Bard, and Llama 2	Private	Human
Falis et al ²²	Synthetic discharge summary generation and ICD-10 coding	GPT-3.5	Public MIMIC-IV	Auto (F1) + Human
Fang et al ²³	Automated taxonomy learning	GPT-4	Public drug labels and proprietary RWD	Auto (accuracy) + Human
Glicksberg et al ²⁴	Predicting admission from emergency department visits	Bio-Clinical-BERT, GPT-4; ensemble with ML, zero-shot, few-shot; RAG	RWD	Auto (AUROC, AUPRC)
Griot et al ²⁵	Medical QA	Mistral 7B; fine-tuning	MedMCQA, MedQA-USMLE, PubMedQA, PMC LitArch textbooks	Auto (eval-harness score) + Human
Hu et al ²⁶	Clinical NER	GPT-3.5, GPT-4	MTSamples, VAERS	Auto (exact and relaxed match precision, recall, F1)
Hua et al ²⁷	Medical QA (TCM—traditional Chinese medicine)	Baichuan2; fine-tuning	Custom TCM datasets	Auto (F1)
Kugic et al ²⁸	Acronym disambiguation	GPT-3.5, GPT-4, Llama 2 (7B-chat, 70B-chat)	CASI (English); 500 text spans (German and Portuguese)	Auto (accuracy)
Li et al ²⁹	Biomedical NER	GPT-4	BC5CDR; NCBI Disease	Auto (F1)
Li et al ³⁰	Literature recommendation and summarization	GPT-4, ScholarAI, Gemini	2 systematic review topics	Human (relevance, quality, etc)
Li et al ³¹	NER and RE	BioBERT, PubMedBERT, BioM-ELECTRA, GPT-3.5, GPT-4; ensemble, zero-shot	LitCoin NLP challenge dataset	Auto (Jaccard Similarity score)
Lucas et al ³²	Medical QA	GPT-3.5, Med42-70B; zero-shot, CoT, ensemble	USMLE	Auto (MCQ accuracy)
Luo et al ³³	Medical QA	Llama 2-70B	RWD	Auto (accuracy)
Nievas et al ³⁴	Synthetic data generation	GPT-3.5, GPT-4, LLaMA (7B, 13B, 70B); CoT, fine-tuning on LLaMA	SIGIR and TREC CT	Auto (NDCG@10, P@10, AUROC, AURPC, Imp. CLA, Exp. CLA)
Peng et al ³⁵	7 clinical NLP tasks: concept extraction, relation extraction, abbreviation disambiguation, natural language inference, medication attribute filling, concept normalization, and progress note understanding	GatorTronGPT	2018 n2c2; 2022 n2c2; SemEval-2015 task 14; UMN abbreviation	Auto (F1)
Platt et al ³⁶	Examination of public comfort with the use of ChatGPT	n/a	Survey of US adults using the NORC AmeriSpeak Panel (n = 1787)	n/a
Remy et al ³⁷	Semantic textual representations	GPT-3.5	12 datasets	Auto (Pearson correlation, Spearman correlation, Top1 accuracy)
Tran et al ³⁸	BioNLP tasks: medical QA; IE and text generation	PMC-LLaMA-7B, Chat-Doctor, LLaMA, Llama 2; instruct tuning	MedQA-USMLE; MedMCQA; Pub-MedQA; BioASQ; MedNLI	Auto (accuracy)
Woo et al ³⁹	Identify actionable, incidental findings from radiology reports	GPT-4; zero-shot	RWD (in-house dataset created from EHRs across 4 academic ED sites)	Auto (F1) + Human
Wu et al ⁴⁰	Medical QA	PMC-LLaMA; fine-tuning	PubMedQA, MedMCQA, USMLE	Auto (Accuracy) + Human
Wu et al ⁴¹	Chinese medical QA	GPT-3.5, GPT-4, Baichuan2 (7B, 13B), QWEN-72B; zero/few-shot	China National Medical Licensing Examination	Auto (Accuracy)

(continued)

Table 1. (continued)

Paper Ref.	Task	Large language models	Dataset	Evaluation (automatic vs human)
Yan et al ⁴²	LLMs to generate executable phenotyping algorithms	GPT-4, GPT-3.5, Claude-2, PaLM 2, Bard	In-house dataset—84 821 local patient cohort.	Auto (TP, FP, PPV, Recall, FPR) + Human
Ying et al ⁴³	Ontology term clustering	Contrastive learning (CoRTE _x)	UMLS	Auto (F1)
Zhang et al ⁴⁴	Disease diagnosis and patient assessment	ChatGPT, GPT-4	MIMIC-III; case reports	Auto (F1) + Human
Zhou et al ⁴⁵	Biomedical relationship extraction	GPT-4, GPT-3.5, Llama 2 (7B, 13B), MedLLa-MA_13B; Fine-tuned “soft prompts” for open-weight LLMs	Public benchmarks (DDI 2013, ChemProt, BioRED)	Auto (F1)
Brief communication articles				
Chen et al ⁴⁶	Symptom-checking for 194 diseases	GPT-4, GPT-3.5 turbo	Mayo Clinic Symptom Checker patient service	Auto (relative comparative score)
Dos Santos et al ⁴⁷	Generation of nursing care plans	GPT-4, GPT-3.5	Available upon request	Human
Kim et al ⁴⁸	Generation of nursing statements	GPT-4, Bio_ClinicalBERT; zero-shot, fine-tuning	Dataset upon request	Auto (AUC, AUPRC) + Human
Pividori and Greene ⁴⁹	AI-assisted academic authoring	GPT-3.5 (davinci, turbo)	5 manuscripts	Auto (GPT-3.5 turbo, GPT-4 turbo) + Human
Vaid et al ⁵⁰	Medical QA (echocardiogram reports)	Llama 2-70B	RWD (in-house dataset created from EHRs)	Auto (Accuracy)
Wang et al ⁵¹	HPO rare disease concept normalization	Llama 2-7B, GPT-3.5; fine-tuning	In-house dataset created based on HPO vocabulary	Auto (accuracy)
Perspective articles				
Benítez et al ⁵²	Role of LLMs in medical education	n/a	n/a	n/a
Hersh ⁵³	Role of LLMs in information retrieval	n/a	n/a	n/a
Rodriguez et al ⁵⁴	Role of LLMs in health equity	n/a	n/a	n/a
Spathis and Kawsar ⁵⁵	Pitfalls in representing and tokenizing temporal data	n/a	n/a	n/a
Review articles				
García Subies et al ⁵⁶	Comparative analysis of Spanish clinical encoder models on NER and classification	BETO, MarIA, Rigo-BERTa, XLM-RoBERTa, DeBERTaV3, GPT family, Galén family, bsc-bio-her-es	17 corpora available in Hugging Face Hub	Auto (F1)
Pool et al ⁵⁷	LLMs in telehealth	n/a	n/a	n/a
Sahoo et al ⁵⁸	LLMs in biomedical and clinical NLP research	n/a	n/a	n/a

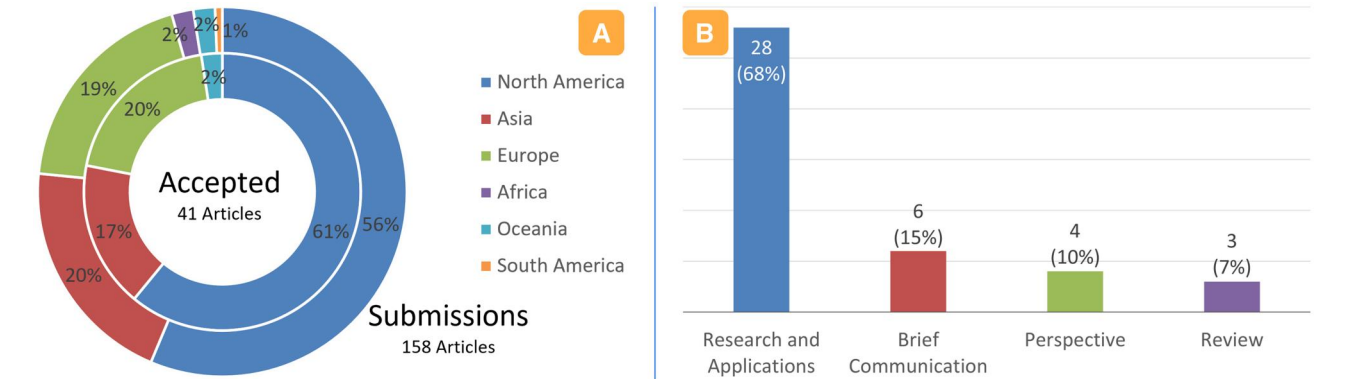


Figure 1. Breakdown of submissions and accepted articles by continent (A) and number and breakdown of accepted articles by content type (B).

While the majority of the accepted articles are regular research papers, we have also received and accepted submissions in the formats of brief communications, perspectives, and reviews. Figure 1B shows a breakdown of the articles by content type.

Key research themes and topics

The papers in the Focus Issue explore a wide range of tasks within two main thematic topics: (a) innovative uses of LLMs in various biomedical and healthcare applications, and (b) technical method development for enhancing the capability of standard LLMs. In the following sections, we briefly describe studies in each of the themes while highlighting five specific papers in more detail.^{33,39,40,45,54}

Biomedical NLP tasks

A number of studies focus on the traditional biomedical NLP tasks such as information extraction, concept normalization, and information retrieval. In the following sections, we highlight the work by Zhou et al⁴⁵ as a representative study in this category.

Information extraction

Information extraction refers to the task of extracting biomedical and clinical concepts and their relations from unstructured text data and turning them into structured knowledge. Such a task is crucial in building knowledge of biomedicine and health.

Zhou et al applied Llama 2-7B, Llama 2-13B, and MedL-LaMA_13B for relation extraction on the drug-drug interaction (DDI),⁵⁹ ChemProt,⁶⁰ and BioRED⁶¹ datasets using their LEAP (LLM instruction-example adaptive prompting) framework to improve performance.⁴⁵ Through benchmarking results, their LEAP approach exhibit superior performance and robustness over conventional instruction tuning.

Given its importance in biomedical NLP research, this task also drew significant attention from many other researchers. Hu et al developed and refined task-specific prompts for GPT-3.5 and GPT-4 to identify clinical concepts such as medical problems, treatments, and tests in the MTSamples⁶² dataset, and clinical events such as investigations, adverse events, and procedures in the VAERS⁶³ dataset.²⁶ Li et al employed GPT-4 for Named Entity Recognition (NER) tasks on the BC5CDR⁶⁴ and NCBI corpora⁶⁵ with a Retrieving and Chain-of-Thought (RT) framework for performance improvement.²⁹ Peng et al adopted GatorTronGPT,⁶⁶ a unified generative clinical LLM, to extract clinical concepts and relations through text-to-text learning formulation and soft prompt tuning.³⁵ Tran et al created a domain-specific instruction dataset and fine-tuned the 7B and 13B LLMs of LLaMA⁶⁷ and Llama 2⁶⁸ on the dataset for medication status extraction and coreference resolution.³⁸

Concept coding and normalization

Concept coding and normalization maps biomedical and clinical concepts in their various expression formats to terms from a controlled vocabulary. Falis et al evaluated using GPT-3.5 for normalization of ICD-10 codes within synthetic discharge summaries generated by GPT-3.5.²² Kugic et al assessed the performance of GPT3.5, GPT-4, Llama 2-7B-Chat, and Llama 2-70B-Chat for disambiguation of acronyms in clinical narratives.²⁸ Yan et al prompted GPT-4, GPT-3.5 of ChatGPT, Claude 2, and Bard to generate executable phenotyping algorithms in the form of SQL queries adhering to a common data model (CDM) for 3 phenotypes, including type 2 diabetes mellitus, dementia, and hypothyroidism.⁴² Wang et al fine-tuned an open source Llama 2 model using a domain-specific corpus sourced from the

Human Phenotype Ontology (HPO) and achieved the current best performance for rare disease concept normalization.⁵¹

Information retrieval

Information retrieval plays an important role in identifying relevant articles from the biomedical literature to support research. Akinseloyin et al proposed a QA framework for using GPT-3.5⁶⁹ to prioritize candidate studies for systematic review.¹⁹ Li et al leveraged GPT-4 turbo¹¹ in their innovative retrieval-augmented generative tool to systematically retrieve literature from PubMed, article recommendation, and summarization.³⁰

Knowledge acquisition

LLMs are pretrained on large amounts of data and have stored lots of statistical patterns about terms and concepts in their memory. These patterns can be distilled from LLMs to accelerate knowledge acquisition and to minimize human effort in this typically laborious and error-prone task. Fang et al integrated LLMs and real-world evidence to automatically develop a taxonomy and demonstrated the effectiveness of this approach through learning a drug indication taxonomy from drug labels to the satisfaction of domain experts.²³

Clinical tasks and applications

A number of studies explored the application of using LLMs as clinical assistants, covering a broad set of topics such as clinical decision support, risk prediction, nursing care, patient trial matching, and AI-assisted publishing. Here, we highlight the work of Woo et al³⁹ as a representative of different studies considering LLMs as future assistants in clinical care.

Clinical decision support

Woo et al evaluated the proficiency of a HIPAA-compliant version of GPT-4 in identifying actionable, incidental findings from unstructured radiology reports of emergency department (ED) patients.³⁹ Real-world radiology reports were first manually reviewed by domain experts for nonemergent, incidental findings with high likelihood of requiring actionable follow-up. GPT-4 were then tuned for optimal performance on a validation set of 50 reports. Their evaluation includes comparing GPT-4 output against the human ground truth in benchmarking experiments. To further validate its results for practical uses, GPT-generated results were also manually reviewed for hallucinations and assessed for appropriateness via Likert scale. Their primary results show that GPT-4 can achieve over approximately 85% accuracy in standard F-1 measure without any hallucinated results. This work demonstrates the promises of AI-generated patient instructions in augmenting diagnostics, as well as highlights the importance of keeping human-in-the-loop when integrating AI/LLM into medical workflow.

Bak and Chin evaluated and highlighted the potential and limitations of using ChatGPT, Bard, Llama 2 in identifying different states of motivation following the TTM (Trans-theoretical Model) stages of change to provide appropriate information for behavior change.²¹ Zhang et al evaluated the performance of ChatGPT, GPT-4, and Llama 2 in identifying patients with specific diseases and their proficiency in facilitating clinical decision-making and supporting patient assessment at various stages in the care pathway.⁴⁴ Vaid et al demonstrated the feasibility and effectiveness of using a

locally employed open-source LLM, namely Llama 2-70B, for querying and interpreting echocardiogram report data.⁵⁰ Chen et al evaluated the accuracy of GPT-3.5, GPT-3.5-turbo, and GPT-4 for symptom-checking across a broad range of diseases.⁴⁶

Risk prediction

Acharya et al fine-tuned the Llama 2-7B model on textual descriptions within structured EHRs for clinical risk prediction.¹⁸ Glicksberg et al assessed the capabilities of GPT-4 for predicting patient admissions from ED triage notes and tabular data.²⁴

Nursing care

Kim et al used GPT-4, Bio_ClinicalBERT⁷⁰ to validate algorithm-generated nursing statements.⁴⁸ Dos Santos et al proposed a framework for creating prompts that facilitate GPT-3.5 and GPT-4 to generate valid nursing care plan suggestions for a hypothetical patient with lung cancer.⁴⁷

Clinical trial matching

Nievas et al used GPT-4 to create a specialized synthetic dataset to fine-tune LLaMA 7B, 13B, 70B and compare the fine-tuned models to GPT-3.5 and GPT-4 on performance for matching patients to clinical trials.³⁴

AI-assisted authoring

Pivadori and Greene developed and evaluated a tool called the Manubot AI Editor as part of a human-centered approach to AI-assisted authoring in which an LLM reviews and suggests revisions to a human-authored paper.⁴⁹

Prompt engineering techniques for enhancing standard LLMs

LLMs are designed to generate plausible text, but they do not always “reference” a reliable source of current facts or justify the factuality of their responses. In the medical domain, such unsubstantiated responses may lead to physical or psychological harm, or undesired changes in therapy or compliance. Additionally, these models are often fine-tuned using human feedback to improve their dialogue abilities, but this process is not foolproof, making incorrect and biased responses inevitable. Due to their previously demonstrated superior performance, the commercially hosted “closed” (proprietary model weights) GPT-3.5 and GPT-4 models and the “open” (shared model weights) Llama 2 model are the most frequently studied among a dozen different models in the Focus Issue (Figure 2A). In addition to showing their performance in the zero-shot learning (instructions without examples) fashion, many studies have also investigated the design and optimization of input prompts for further performance enhancements. The three commonly used techniques are given below:

Few-shot learning

Few-shot learning refers to the use of a few examples (eg, in addition to instructions) in prompt inputs, allowing LLM models to generalize from limited data. For instance, Wu et al proposed using few-shot learning from past medical questions as part of the Knowledge and Few-shot Enhancement in-context learning framework (that also integrates external clinical knowledge from a knowledge base of Chinese medical books) to enhance the performance of GPT-3.5, GPT-4,

Baichuan2-7B, Baichuan2-13B,⁷¹ and QWEN-72B⁷² on CNMLE-2022.^{41,73}

Chain-of-thought (CoT) prompting

Chain-of-thought (CoT) prompting involves designing prompts that lead the model through a step-by-step reasoning process. For example, Lucas et al proposed an ensemble reasoning prompting approach to improve the performance of GPT-3.5 turbo, GPT-4 turbo, and Med42-70B on medical question answering on questions from the United States Medical Licensing Examination (USMLE).³² Nievas et al employed a CoT reasoning approach to several LLMs, including the LLaMA 7B, 13B, and 70B models fine-tuned on a specialized synthetic dataset generated by GPT-4, as well as GPT-3.5 and GPT-4 itself to produce a step-by-step explanation for matching patients to clinical trials.³⁴

Retrieval-augmented generation

Retrieval-augmented generation (RAG) is a technique that combines retrieval-based and generation-based methods to improve the performance of language models. For example, the prompt provided to a model may be augmented with relevant text from an authoritative source. As one example, Li et al proposed an innovative retrieval-augmented generative tool that leverages GPT-4 turbo for systematic literature retrieval from PubMed, article recommendation, and summarization.³⁰ As another example, Glicksberg et al conducted experiments using RAG to retrieve the 10 most relevant historical cases to augment GPT-4’s decision-making process of predicting patient admissions from ED visits, outperforming GPT-4 without RAG augmentation.²⁴

Creating domain-/language-specific LLMs

LLMs, such as GPT-4, have demonstrated significant capabilities across a wide range of general-domain tasks, leveraging vast amounts of data to understand and generate human-like text. However, these models often exhibit gaps in specialized knowledge areas or domains, particularly in the medical domain, where targeted expertise and precision are critical. Domain-specific LLMs have emerged to address these deficiencies by pretraining (or fine-tuning) on narrowly focused datasets to develop a deep understanding of domain-specific languages and nuances, as exemplified in multiple studies in the Focus Issue. In the sections below, we highlight PMC-LLaMA (English) and Taiyi (Bilingual) as two representative studies in this category,^{33,40} along with brief description of other relevant works.

Model fine-tuning

Several papers concern the development and/or application of a locally developed domain-specific LLM made available to the research community. Wu et al introduced two PMC-LLaMA models derived by further fine-tuning differently sized Llama 2 models on a curated dataset, including biomedical textbooks and academic literature, as well as medically focused sets of instructions with ideal responses, with improvements in performance on a range of benchmark tasks.⁴⁰ As one of the first efforts in the biomedical domain, their work demonstrates the feasibility and potential benefits of creating domain-specific foundation models and contributes a large-scale, comprehensive training dataset for instruction tuning.

Sahoo et al characterized the opportunities, challenges, and best practices of using LLMs in biomedical and clinical NLP tasks.⁵⁸

Gaps, opportunities, and future directions

LLM evaluation beyond multiple-choice questions

As shown in Table 1, multiple-choice questions (MCQs) remain as a popular form of evaluation in this Focus Issue. While being straightforward and efficient, such an evaluation method does not reflect real-world scenarios where potential answers are not provided ahead of time. Previous research has shown that LLM performance drops significantly when testing moves from MCQs to open-form evaluations.⁷⁶ Moreover, automated evaluation is more commonly used in the included studies as compared with manual assessment of model output (Figure 2B). Common evaluation metrics, such as F1 and area under the curve (AUC), provide quantitative measures of performance for tasks such as concept recognition, information extraction, and retrieval. However, they are not designed to evaluate the performance of text generation, which is crucial for LLMs. On the other hand, widely used metrics like BLEU and ROUGE have shown some consistency with human judgment on machine translation and summarization tasks, respectively. Still, they depend upon lexical overlap (ie, word matching) and evaluate neither the semantics (the meaning underlying the words) nor the clinical utility or veracity required for AI deployment in healthcare. These challenges have led to a growing focus on comprehensive human evaluations to ensure LLMs are reliable, trustworthy, safe, and ethical for use in real-world healthcare settings. As such, it is encouraging to see several studies in the Focus Issue that employed manual expert review to assess the quality of LLMs generated output beyond traditional accuracy metrics.^{20–23,25,30,39,40,42,44,47,48}

Multimodal LLMs

LLMs studied in this Focus Issue have made great strides in several areas, including QA, information extraction, and summarization. However, they have predominantly centered on the text-only data modality—partly because the accepted papers reflect work prior to the release of recent models such as GPT-4V⁷⁷ and Gemini.⁷⁸ In the healthcare field, the development of multimodal LLM (MLLM) is pivotal, owing to their ability to transform healthcare by synthesizing information from diverse data sources such as biosensors, genetic, epigenetic, proteomic, microbiome, metabolomic, imaging, clinical, social, and environmental variables. These models could pave the way towards personalized medicine, real-time pandemic surveillance, digital clinical trials, and virtual health coaching, offering a holistic view of an individual's health status.

In the general domain, numerous approaches have been proposed to extend LLMs to multimodal data.^{79,80} For instance, one could employ LLM as a processing unit capable of reasoning based on multimodal information. Armed with powerful LLMs, multimodal LLMs are anticipated to tackle more complex tasks than possible with unimodal models (eg, interpreting radiological images and reports together). Nonetheless, such model developments in the healthcare sector have been relatively slow, despite a few early attempts to evaluate generalist LLMs in biomedicine.^{81–83} This is largely

due to the vast and multidimensional nature of medical data, as well as the lack of data available to model developers. Consequently, it is imperative to identify and address the barriers to biomedical multimodal data analysis and formulate and assess AI solutions capable of effectively harnessing such data.

Safety, fairness, and accountability

A few studies in the Focus Issue have explored ethical issues such as privacy preservation,⁵⁰ the potential bias of demographic data,²⁰ public comfort,³⁶ and the opportunities and challenges of using LLMs in supporting health equity.⁵⁴ However, one significant gap is the current lack of robust approaches and mechanisms to ensure the accuracy and safety of the information provided by LLMs, which is critical in healthcare decisions. Moreover, these models often lack transparency in their decision-making processes and training data, which can lead to challenges in gaining trust from healthcare professionals and patients, and make the causes of model error difficult to ascertain. However, the potential for LLMs to improve healthcare quality, efficiency, and safety by processing vast amounts of data for diagnostics, personalized medicine, and patient management is immense. Future research should focus on developing ethical frameworks and technical standards that enhance LLMs' safety, reliability, fairness, and accountability.^{84,85} By addressing these needs, LLMs can be more responsibly integrated into healthcare systems, ultimately enabling improved patient outcomes and operational efficiencies.

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Author contributions

Trevor Cohen, Zhiyong Lu, Marzyeh Ghassemi, Yifan Peng, and Chunhua Weng managed the review of the manuscripts for the Focus Issue. All authors contributed to the drafting of the editorial and approved its final version.

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Conflicts of interest

All authors declare no conflict of interests.

Data availability

No new data were generated or analyzed in support of this research.

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